Expt No: 9

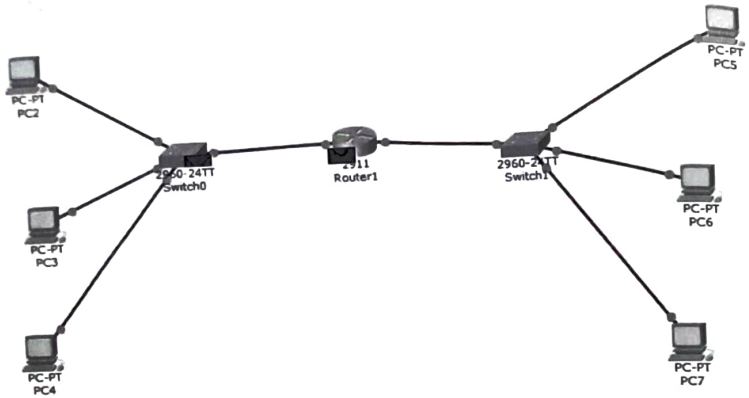
Date: 20/8/25

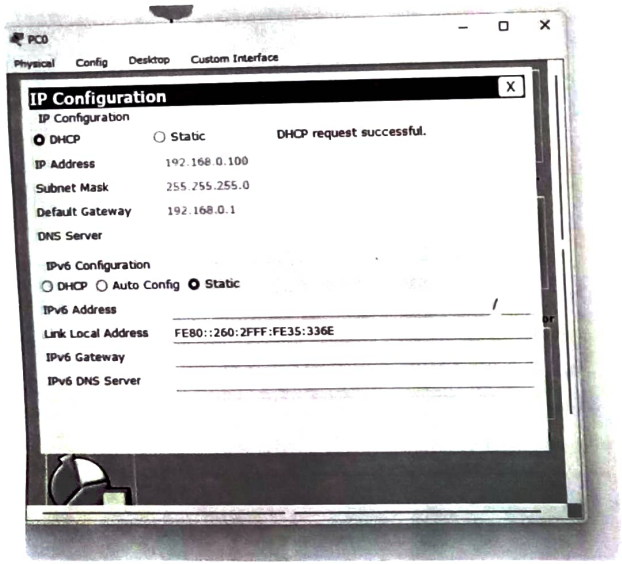
Implementation of SUBNETTING In CISCO PACKET TRACER Simulator

a) Design Multiple Subnets with a Suitable Number of Hosts

Aim:

To implement Subnetting by individual a large network into multiple Smaller Subnetworks according to the required number of hosts and to Study the allocation of IP addresses in Cisco packet Tracer.

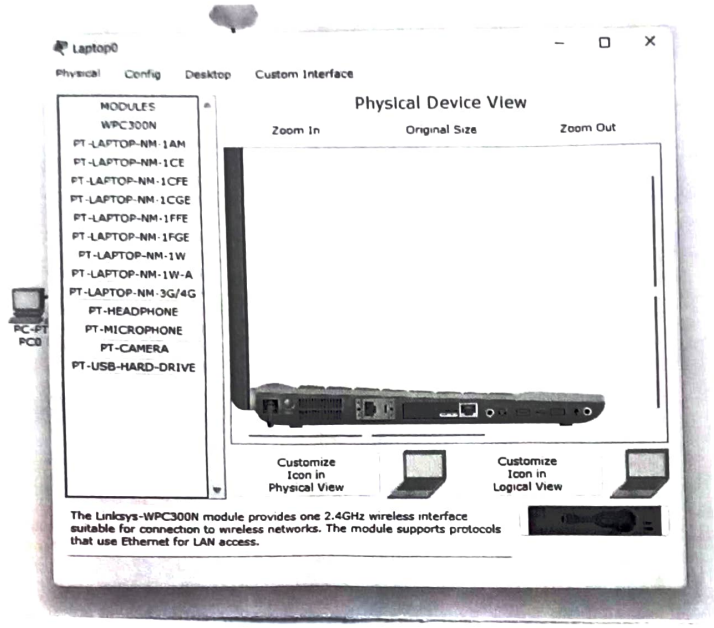


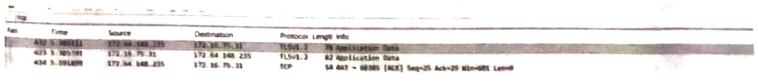


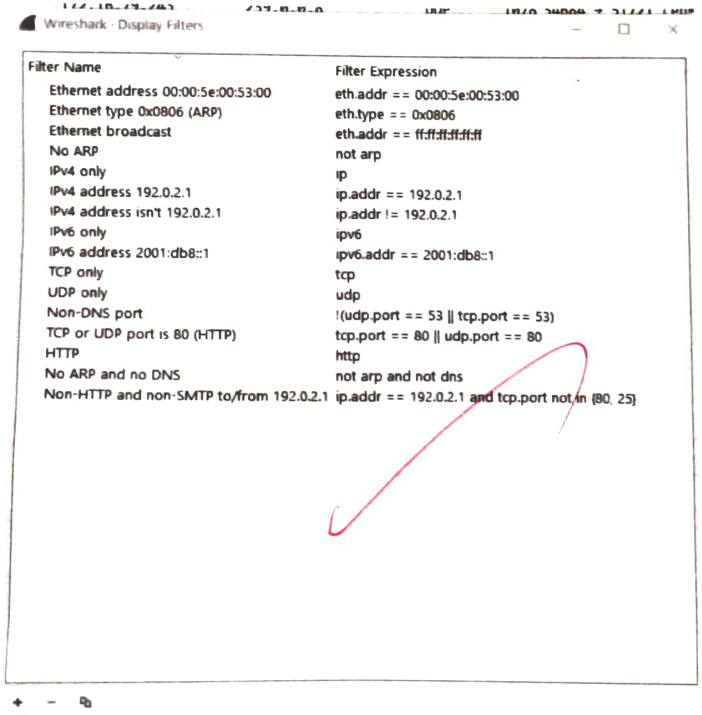
Steps:-

1. Open Cisco packet Tracer and create a new project.
2. Design the network topology by placing the required network devices such as routers, Switches and PCs.
3. Select a Suitable network address for the experiment.
4. Calculate the number of Subnets required.
5. Determine the number of hosts needed in each Subnet
6. Calculate the Subnet mask using Subnetting techniques
7. Divide the network into multiple Subnets.
8. Assign a Seperate networks address to each Subnet.
9. Verify the Subnet Calculations and host ranges
10. Save the network topology

Result: Thus, the multiple Subnets were Successfully created according to the required number of hosts, and the network address was efficiently divided into Smaller Subnets.







Expt 9(b) Assign Static IP Address Across All Subnets and Connect the Subnets Through a Router.

Aim: To assign Static IP Address to all devices in different Subnets and Establish Communication between the Subnets through a router.

Steps:

1. place a router in the network topology.
2. Connect the router to the Switches using appropriate Cables.
3. Configure the router interfaces.
4. Assign and IP address and Subnet mask to each router interfa
5. Configure the Switches.
6. Assign Static IP address to all Computers in every Subne
7. Configure the Subnet mask for each devices.
8. Assign the appropiate default getaway to each Computer.
9. Verify the IP Configuration of all devices.
10. Save the router Configuration.
11. Check the network topology for Configuration errors.
12. Verify that all devices are properly Connected

Results:-

Thus, Static IP address were Successfully assigned all devices and the Communication between different Subnets was established through the router.

Expt (9c) Simulate packet Transmission Across the Subnets and Observe the Result

(i) When Subnets Are Connected through a Router

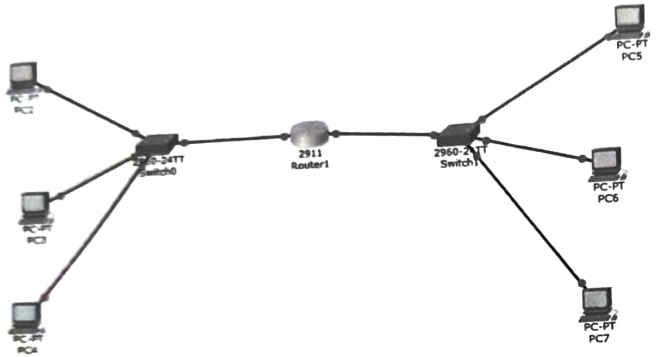
Aim:- To analyze packet Transmission between different Subnets when a router is used for communication.

Steps:

1. Open Simulation Mode in Cisco Packet Tracer.
2. Select a Source devices from one Subnet.
3. Select a destination devices from another Subnet.
4. Generate a packet using the Simple PDU tool.
5. Observe the movement of the packet through the network.
6. Analyze how the router receive the packet.
7. Observe the forwarding process performed by the router.
8. Verify the packet transmission between different Subnets.
9. Use the ping command to test Connectivity.
10. Observe the Successful delivery of packets.
11. Record the Simulation results.
12. Analyze the routing process.

Result:

Thus, packets were Successfully transmitted between different Subnets through the router. The router received the packet from the Source Subnet, processed them and forward them to the destination Subnet. Successful ping response Confirmed paper Commication between all network devices.




```
--- System Configuration Dialog ---

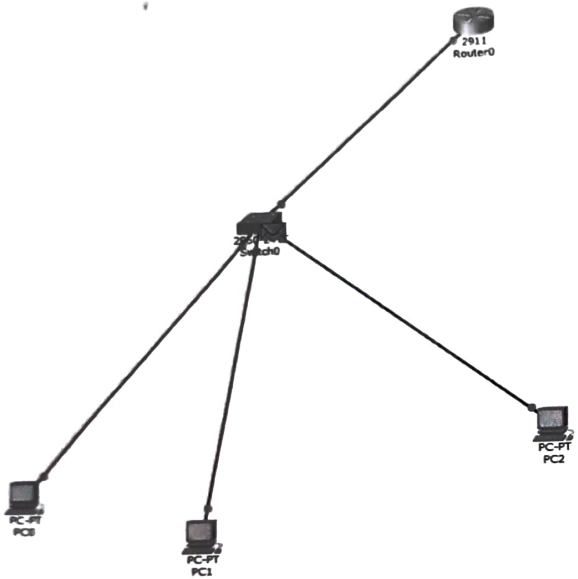
Continue with configuration dialog? [yes/no]: no

Press RETURN to get started!


Router>enable
Router$configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)$interface gigabitethernet 0/0
Router(config-if)$ip address 192.168.10.1 255.255.255.0
Router(config-if)$no shutdown

Router(config-if)$
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state t

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEtherne
to up
exit
```



(ii) When Subnets Are Not Connected Through a Router

Aim:- To analyze packet transmission between different Subnets when no router is available.

Steps:

1. Remove the router from the network topology.
2. Open Simulation Mode.
3. Select devices belonging to different Subnets.
4. Generate packets using the Simple PDU tools.
5. Observe the packet transmission.
6. Perform a ping test between devices located in different Subnets.
7. Analyze the packet flow.
8. Observe the Simulation events.
9. Identify the point at which packet transmission fails.
10. Compare the result with the network that uses a router
11. Record the Observations.
12. Analyze the importance of a router in inter-network communication.

Result:

Thus, Communication between devices located in different Subnets failed when a router was not used. The packets could not reach the destination because there was no routing device to forward them. This experiment demonstrated the important of routers in enabling communication between